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LUMINA PROTOCOL

On-Chain Parametric Insurance for Artificial Intelligence Agents

Network: Base L2 (Chain 8453)

Settlement: USDC (Circle)

Yield: Aave V3

Version: 3.0 -- March 2026

Contact: hello@lumina-org.com

Docs: <https://lumina-org.com>

Base L2

USDC

Aave V3

Parametric

AI Agents

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1. Executive Summary

Lumina Protocol is the first decentralized parametric insurance infrastructure designed exclusively for artificial intelligence agents operating in DeFi. The protocol deploys on Base L2, settles in real USDC (Circle), and generates yield through Aave V3.

Unlike traditional insurance, which requires a human to file a claim, a committee to review it, and weeks of waiting to receive payment, Lumina uses mathematical triggers verified by oracles. If the condition is met (for example, ETH drops 30%), the payout is instant and automatic. No claims. No disputes. No waiting for humans.

The protocol offers **5 insurance products** (BTC Catastrophe Shield, ETH Apocalypse Shield, Depeg Shield, IL Index Cover, Exploit Shield), **4 liquidity vaults** (VolatileShort, VolatileLong, StableShort, StableLong) and operates through **13 smart contracts** deployed in production with UUPS upgradeable proxies.

Every interaction flow begins the same way: **the human, through their AI agent**, instructs the desired operation. The agent executes the on-chain transaction autonomously, interacting with Lumina's API and protocol contracts without manual intervention.

The business model is simple and transparent: a 3% fee on the premium paid by the insured, a 3% fee on the claim payout, and a 3% performance fee on the positive yield (profit) when liquidity providers withdraw funds from vaults.

2. Introduction

2.1 The Problem: DeFi Without Protection for Autonomous Agents

The agentic economy is growing exponentially. Thousands of AI agents manage DeFi positions, treasuries, liquidity pools, and yield farming strategies. However, these agents operate without any type of coverage against the most critical risks in the ecosystem: catastrophic market crashes, stablecoin depegs, severe impermanent loss, and protocol exploits.

Existing DeFi insurance protocols were designed for humans. They require graphical interfaces, community voting processes to validate claims, and resolution times incompatible with the speed at which autonomous agents operate. Lumina solves this problem with a purely programmatic and parametric approach.

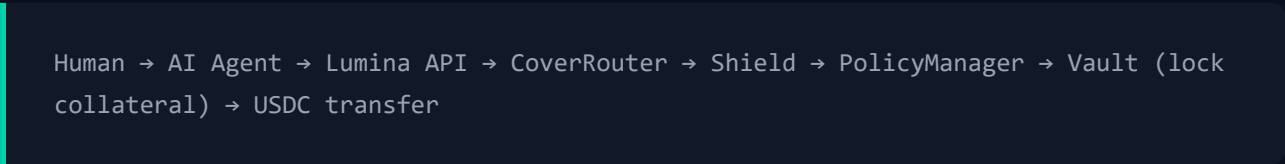
2.2 Comparison with Existing Protocols

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Primary focus	Insurance for humans	Multi-chain insurance	Parametric insurance	Insurance for AI agents
Insurance type	Discretionary	Discretionary	Parametric	Parametric
Claims process	Community voting	Evaluation committee	Automatic oracle	Automatic oracle
Payment time	Weeks	Days	Hours	Minutes (1 tx)
Designed for AI agents	No	No	Partial	Yes, exclusively
Yield for LPs	Staking NXM	Farming	Premiums	Aave V3 + Premiums
Collateralization	Shared pool	Shared pool	Per risk	1:1 per policy
Network	Ethereum L1	Multi-chain	Polygon	Base L2
Settlement	ETH/DAI	Multi-token	USDC	USDC
Products	Smart Contract Cover	Various	Weather/Parametric	4 DeFi products

2.3 How Lumina Works

The human, through their AI agent, defines the desired coverage parameters: asset to cover, amount, duration, and product. The agent queries Lumina's API to obtain a precise quote, including the premium calculated using the kink utilization model. Once the quote is approved, the agent executes the on-chain transaction through the CoverRouter.

Simplified flow:



The collateral is locked 1:1 in the corresponding vault. If the trigger is activated during the policy's validity period, the payout is executed automatically. If the policy expires without the trigger being activated, the collateral is released back to the vault.

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3. Protocol Architecture

3.1 Kink Utilization Model

The core of Lumina's pricing is a non-linear utilization model inspired by the interest rate models of Aave and Compound. The premium is calculated with the following complete formula extracted from the Solidity code:

$$\text{Premium} = \text{Coverage} \times \text{P_base} \times \text{RiskMult} \times \text{DurationDiscount} \times \text{M(U)} \times (\text{Duration} / 365)$$

Where:

- **Coverage:** insured amount in USDC
- **P_base:** annualized base rate for the product (e.g.: 15% for BCS, 20% for EAS, 2.5% for Depeg)
- **RiskMult:** risk multiplier per asset
- **DurationDiscount:** discount for long duration
- **M(U):** utilization multiplier (kink model)
- **Duration:** policy duration in days

Kink Model -- Utilization Multiplier M(U)

The inflection point (kink) is located at $U_{\text{kink}} = 80\%$. Below the kink, the multiplier grows linearly. Above it, it grows aggressively to disincentivize overutilization.

```
If U <= U_kink (80%): M(U) = 1 + (U / U_kink) x 0.5 If U > U_kink (80%): M(U) = 1 + 0.5 + ((U - 0.8) / (1 - 0.8)) x 3.0 // U_MAX = 95% → policy is rejected (coverage cannot be purchased)
```

Multiplier Table by Utilization Level

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0%	1.00x	Base premium	
20%	1.13x	+13% over base premium	
40%	1.25x	+25% over base premium	
60%	1.38x	+38% over base premium	
80%	1.50x	+50% (kink point)	
85%	2.25x	+125% (aggressive zone)	
90%	2.25x	+125% (aggressive zone)	
95%	3.75x	REJECTED (U_MAX reached)	

This model ensures that when the vault capacity is ample, premiums are competitive. As utilization approaches 80%, premiums rise gradually. Above 80%, growth is exponential, protecting LPs from overexposure.

Vault Investment Model

LPs deposit USDC into Lumina's vaults. This USDC is automatically deposited into Aave V3 to generate a base yield (3-5% APY). Additionally, insurance premiums paid by policy buyers flow to the vault, increasing the value of LP shares.

The total yield for LPs is composed of:

- **Aave V3 base yield:** 3-5% APY generated by the USDC deposit in Aave V3
- **Insurance premium yield:** Premiums paid by policy buyers (97% after the protocol fee) are distributed to the vault
- **Total yield = Aave V3 base yield + insurance premium yield**

The Kink Model also protects LPs: the higher the vault utilization, the higher the premiums charged, which translates to higher yield for liquidity providers. This creates a natural equilibrium where LPs are compensated proportionally to the risk they assume.

3.2 Policy Purchase Flow

The human, through their AI agent, initiates the coverage purchase. The complete on-chain flow is:

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1. **Human** instructs their AI agent with the desired parameters
2. **AI Agent** calls Lumina's API to obtain a quote
3. **API** calculates the premium using the kink model and returns the parameters
4. **Agent** approves USDC to the CoverRouter and executes `purchaseCover()`
5. **CoverRouter** (`0xd5f8678A0F2149B6342F9014CCe6d743234Ca025`) validates parameters and routes to the correct Shield
6. **Shield** (BCS/EAS/Depeg/IL/Exploit) validates product-specific rules
7. **PolicyManager** (`0xCCA07e0676222AA27DEd58482DeD3d9a7d0162a`) registers the policy on-chain
8. **Vault** locks collateral 1:1 to back the coverage
9. **USDC** is transferred from the agent to the vault (premium) and the 3% fee is charged

3.3 Liquidity Provider (LP) Flow

The human, through their AI agent, deposits USDC into one of the four vaults to earn yield:

1. **Human** decides to deposit in a vault and communicates it to their agent
2. **Agent** approves USDC and calls `deposit()` on the selected vault
3. **Vault** receives the USDC and deposits it into **Aave V3** (`0xA238Dd80C259a72e81d7e4664a9801593F98d1c5`)
4. The LP receives **soulbound shares** (non-transferable) proportional to their deposit
5. The shares accumulate yield from two sources: Aave V3 base rate (3-5% APY) + insurance premiums
6. To withdraw, the LP initiates a **cooldown** period (37 to 372 days depending on the vault)
7. After the cooldown, the LP can execute `withdraw()` and receive their USDC + accumulated yield
8. **A 3% performance fee is charged on the positive yield (profit) upon withdrawal**

3.4 Contract Architecture Table

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CoverRouter	Core	UUPS	All Shields, PolicyManager	0xd5f8...a025
PolicyManager	Core	UUPS	Vaults, Shields	0xCCA0...162a
LuminaOracle	Oracle	UUPS	Chainlink feeds, Sequencer	0xB52B...4C09
LuminaPhalaVerifier	Oracle	UUPS	Phala TEE network	0x468b...9B
VolatileShort Vault	Vault	UUPS	Aave V3, USDC	0xbd44...880d
VolatileLong Vault	Vault	UUPS	Aave V3, USDC	0xFee5...f904
StableShort Vault	Vault	UUPS	Aave V3, USDC	0x429b...54fC
StableLong Vault	Vault	UUPS	Aave V3, USDC	0x1778...321c
BTC Catastrophe Shield / ETH Apocalypse Shield	Product	UUPS	Oracle, PolicyManager, VolatileShort	0x54CD...844e
DepegShield	Product	UUPS	Oracle, PolicyManager, StableShort	0x71DB...8306
ILIndexCover	Product	UUPS	Oracle, PolicyManager, VolatileShort	0x4196...4E0
ExploitShield	Product	UUPS	Oracle, Phala TEE, PolicyManager	0xAE29...b38C
TimelockController	Governance	--	Gnosis Safe	0xd0De...747a

3.5 Strict 1:1 Collateralization

Every policy issued on Lumina is backed by collateral locked 1:1 in the corresponding vault. This means that if an agent purchases \$50,000 of BCS coverage, the vault locks exactly \$50,000 in USDC (deposited in Aave V3 as aUSDC) to guarantee the payout in case of a claim.

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This model eliminates the undercapitalization risk that affects other DeFi insurance protocols operating with shared pool models. If vault utilization reaches 95% (U_MAX), no new policies are accepted until capacity is freed.

The 1:1 locked collateral does not remain idle. It is automatically deposited in Aave V3 where it generates base yield (3-5% APY) while remaining available to cover claims. This means that even capital committed to active policies is generating yield for LPs.

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4. Insurance Products

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4.1 BTC Catastrophe Shield (BCS)

Through their AI agent, users can protect their BTC positions against catastrophic market crashes. BTC Catastrophe Shield covers extreme scenarios: drops greater than 50% like the one during COVID (March 2020). Events like LUNA (May 2022, BTC -42%) or FTX (November 2022, BTC -26%) do NOT trigger this product because BTC drops were less than 50%.

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Value

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Product ID	BTCCAT-001
Contract	0x36e37899D9D89bf367FA66da6e3CebC726Df4ce8
Trigger	Drop > 50% from purchase price
TRIGGER_DROP_BPS	5000 (50% in basis points)
Verification	TWAP 15 minutes or 3 consecutive Chainlink rounds
Deductible	20%
Payout	Binary: 80% of coverage
Duration	7 to 30 days
Waiting period	1 hour (anti-front-running)
Covered assets	BTC only
MAX_PROOF_AGE	30 minutes
Base rate	15% annualized (1500 BPS)
Max allocation	30% of vault (3000 BPS)
Vault	VolatileShort (37 days cooldown)
Historical frequency	~0.125 events/year (1 in 8 years)
Reference event	COVID March 2020: BTC \$9,100 → \$3,800 (-58%)

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4.1b ETH Apocalypse Shield (EAS)

Through their AI agent, users can protect their ETH positions against apocalyptic market crashes. ETH Apocalypse Shield covers the most extreme scenarios: drops greater than 60%. Only one event of this magnitude has occurred in the last 8 years (COVID, March 2020), when ETH dropped from \$230 to \$80 in 2 days. Events like China (May 2021, ETH -56%) or LUNA (June 2022, ETH -51%) do NOT trigger this product.

Parameter	Value
Product ID	ETHAPOC-001
Contract	0xA755D134a0b2758E9b397E11E7132a243f672A3D
Trigger	Drop > 60% from purchase price
TRIGGER_DROP_BPS	6000 (60% in basis points)
Verification	TWAP 15 minutes or 3 consecutive Chainlink rounds
Deductible	20%
Payout	Binary: 80% of coverage
Duration	7 to 30 days
Waiting period	1 hour (anti-front-running)
Covered assets	ETH only
MAX_PROOF_AGE	30 minutes
Base rate	20% annualized (2000 BPS)
Max allocation	25% of vault (2500 BPS)
Vault	VolatileLong (97 days cooldown)
Historical frequency	~0.125 events/year (1 in 8 years)
Reference event	COVID March 2020: ETH \$230 → \$80 (-65%)

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Circuit Breaker
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The contract implements a circuit breaker mechanism to protect the protocol during periods of extreme volatility:

- Drop > 5% in 1 hour: the premium is multiplied by 1.5x
- Drop > 10% in 1 hour: HALT is activated -- no new policies are accepted

Payout Example

Coverage: \$50,000 in ETH Price at purchase: \$2,000 Trigger price: $\$2,000 \times 0.70 = \$1,400$ ETH drops to \$1,350 → Trigger activated Gross payout: $\$50,000 \times 80\% = \$40,000$ Protocol fee (3%): \$1,200 Net payout to agent: **\$38,800**

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4.2 Depeg Shield

The human, through their AI agent, can protect their stablecoin positions against loss of peg. Depeg Shield covers the scenario in which a stablecoin falls below \$0.95, as occurred with USDC during the SVB crisis (March 2023, reaching \$0.87).

Parameter	Value
Product ID	DEPEG-STABLE-001
Contract	0x7578816a803d293bbb4dbea0efbed872842679d0
Trigger	Stablecoin price < \$0.95
TRIGGER_PRICE	95_000_000 (8 Chainlink decimals)
Verification	TWAP 30 minutes or 5 consecutive rounds
Duration	14 to 365 days
Waiting period	24 hours
Base rate	24% annualized
Vault (short)	StableShort (0x429b...54fC)
Vault (long)	StableLong (0x1778...321c)

Deductibles and Payouts by Stablecoin

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	Deductible	Payout	Net Payout (post 3% fee)	Note
DAI	12%	88% binary	85.36% of coverage	MakerDAO, moderate risk
USDT	15%	85% binary	82.45% of coverage	Tether, centralized risk
USDC	--	EXCLUDED	--	Circular: Lumina settles in USDC

USDC is excluded from coverage because it would create a circular dependency: if USDC loses its peg, Lumina's payouts (which are made in USDC) would also lose value.

Duration Discount

Duration Range	Discount Factor	Effect
14 - 90 days	1.00x	No discount
91 - 180 days	0.90x	10% discount on premium
181 - 365 days	0.80x	20% discount on premium

Duration discounts incentivize long-term policies, which benefits LPs by providing more predictable premium flows.

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4.3 IL Index Cover

The human, through their AI agent, can protect their liquidity provider positions against impermanent loss. IL Index Cover uses the standard Uniswap V2 formula (50/50 pool) to calculate IL precisely and on-chain, with proportional payout based on the IL incurred.

Parameter	Value
Product ID	IL-INDEX-001
Contract	0x2ac0d2a9889a8a4143727a0240de3fed4650dd93
Trigger	IL > 2% at policy expiration
Style	European-style (resolution only at expiration)
Resolution window	48 hours post-expiration
Deductible	2% (ratchet -- only covers IL above 2%)
Payout factor	90%
IL cap	13% (maximum payout = 11.7% of coverage)
Duration	14 to 90 days
Vault	VolatileShort (0xbd44...880d)

Impermanent Loss Formula (ILMath.sol)

```
// r = priceExpiry / pricePurchase // IL = 1 - (2 * sqrt(r)) / (1 + r) //
Payout calculation: ilNet = max(0, IL% - 2%) // Ratchet deductible
rawPayout = coverage x ilNet x 0.90 // Payout factor 90%
maxPayout = coverage x 13% x 0.90 // = 11.7% of coverage
payout = min(rawPayout, maxPayout)
```

The payout is proportional: the greater the net IL (after the deductible), the higher the payment, up to the cap of 11.7% of coverage.

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Impermanent Loss Reference Table (ILMath.sol)

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Price Change	Price Ratio	Gross IL	Net IL (2% ded)	Payout (\$50K, 90%)
+/-10%	0.90 / 1.10	0.14%	0%	\$0
+/-20%	0.80 / 1.20	0.56%	0%	\$0
+/-22%	0.78 / 1.22	0.68%	0%	\$0
+/-25%	0.75 / 1.25	1.03%	0%	\$0
+/-30%	0.70 / 1.30	1.57%	0%	\$0
+/-35%	0.65 / 1.35	2.22%	0.22%	\$99
+/-40%	0.60 / 1.40	3.02%	1.02%	\$459
+/-50%	0.50 / 1.50	5.72%	3.72%	\$1,674
+/-60%	0.40 / 1.60	9.27%	7.27%	\$3,272
+/-75%	0.25 / 1.75	18.35%	13%+ (capped)	\$5,850 (max)
+/-80%	0.20 / 1.80	22.54%	13%+ (capped)	\$5,850 (max)

The maximum absolute payout is **\$5,850** per **\$50,000** of coverage (11.7% of coverage). IL is symmetric: a 50% increase and a 50% decrease produce the same IL (5.72%).

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4.4 Exploit Shield

The human, through their AI agent, can protect their deposits in DeFi protocols against exploits, hacks, and smart contract vulnerabilities. Exploit Shield uses a dual trigger system to minimize false positives.

Parameter	Value
Product ID	EXPLOIT-SHIELD-001
Contract	0x9870830c615d1b9c53dfee4136c4792de395b7a1
Dual trigger	(1) Governance token -25% in 24h AND (2) Receipt token -30% for 4h OR contract paused
Verification	Oracle (Chainlink) + Phala TEE
Deductible	10%
Payout	Binary: 90% of coverage
Cap per wallet	\$50,000
Duration	90 to 365 days
Waiting period	14 days
Vault	StableLong (0x1778...321c)

Dual Trigger System

The dual trigger requires that **both conditions** are met simultaneously:

1. **Primary trigger (Oracle):** The governance token of the covered protocol must have dropped at least 25% in the last 24 hours, verified by Chainlink.
2. **Secondary trigger (TEE):** At least one of these conditions: (a) the protocol's receipt token has dropped 30% or more for at least 4 consecutive hours, or (b) the protocol's main contract has been paused.

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The secondary trigger verification is performed by the Phala TEE (Trusted Execution Environment) at `0x468b5b2f9e43c304c78d386c290b098ae23ad0e`, providing an additional layer

of security against oracle manipulation.

Covered Protocols

Protocol	Tier	Base Rate	Notes
Compound III	Tier 1	Lower	Extensively audited, low risk
Uniswap V3	Tier 1	Lower	Immutable, low risk
MakerDAO	Tier 1	Lower	Robust governance
Curve	Tier 2	Higher	Greater complexity, moderate risk
Morpho	Tier 2	Higher	Newer protocol, moderate risk
Aave V3	--	EXCLUDED	Circular: Lumina deposits in Aave V3

Aave V3 is excluded because Lumina deposits vault funds into Aave V3 to generate yield. Covering an Aave V3 exploit would create a circular dependency: if Aave V3 is hacked, the funds to pay the claim would be compromised.

5. Dynamic Pricing Model (Kink Model)

5.1 Foundations

The Kink Model is a dynamic pricing system inspired by Aave and Compound's interest rate models, adapted for parametric insurance. Its dual objective is to guarantee permanent vault solvency and create a self-balancing market where supply (LP liquidity) and demand (agent policy purchases) find equilibrium naturally.

Unlike a fixed price, the Kink Model adjusts premium costs in real-time based on available capital in each vault. When liquidity is abundant, premiums are affordable. When capital is scarce, premiums increase dramatically, protecting LPs and attracting new capital.

5.2 Calculation Formula

The complete formula, extracted from PremiumMath.sol:

```
Premium = Coverage x P_base x RiskMult x DurationDiscount x M(U) x (Duration / 365)
```

Where M(U) is the utilization multiplier:

```
If U <= U_kink (80%): M(U) = 1 + (U / U_kink) x R_slope1
M(U) = 1 + (U / 0.80) x 0.5
If U > U_kink (80%): M(U) = 1 + R_slope1 + ((U - U_kink) / (1 - U_kink)) x R_slope2
M(U) = 1 + 0.5 + ((U - 0.80) / 0.20) x 3.0 // If U > 95%: Policy REJECTED
```

On-chain parameters (PremiumMath.sol):

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Kink Point	80%	<code>U_KINK = 8000 bps</code>	
Pre-kink slope	0.5	<code>R_SLOPE1_WAD = 5e17</code>	
Post-kink slope	3.0	<code>R_SLOPE2_WAD = 3e18</code>	
Max utilization	95%	<code>U_MAX = 9500 bps</code>	
Internal precision	1e18	<code>WAD</code>	

Multiplier table:

Utilization	M(U)	Effect
0%	1.00x	Base premium
20%	1.13x	+13%
40%	1.25x	+25%
60%	1.38x	+38%
80%	1.50x	+50% (kink point)
85%	2.25x	+125% (aggressive zone)
90%	2.25x	+125%
95%	3.75x	REJECTED

Numerical example: An agent buys BTC Catastrophe Shield for \$50,000 coverage, 14 days, with U=50%:

```
P_base = 1500 bps (15% annualized) RiskMult = 1.0x (ETH base) DurationDiscount = 1.0x
M(U) = 1 + (0.50/0.80) x 0.5 = 1.3125x Premium = 50,000 x 0.15 x 1.0 x 1.0 x 1.3125 x (14/365) = $377.57
```

Same scenario with U=85%:

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$$((0.85-0.80)/0.20) \times 3.0 = 2.25 \times \text{Premium} = 50,000 \times 0.1$$

$$1.0 \times 2.25 \times (14/365) = 3647.26$$

5.3 Vault Utilization

Utilization (U) is defined as:

$$U = \text{Capital locked in active policies} / \text{Total vault assets}$$

- "Locked capital" = USDC reserved 1:1 to cover active policies (`allocatedAssets` in BaseVault.sol)
- "Total assets" = aToken balance in Aave V3 + free USDC in vault (`totalAssets()` in BaseVault.sol)

When an agent purchases a policy, corresponding capital is locked. When the policy expires or executes, capital is released.

5.4 Self-Balancing Mechanism

The Kink Model creates a market that self-balances through capital scarcity:

When insurance demand is high and vaults have little available liquidity:

- Premiums rise, making coverage expensive for agents
- LP yields increase, rewarding them for higher risk
- New LPs are attracted by high returns, injecting more capital
- Utilization drops, premiums normalize — the cycle repeats

When there is abundant liquidity and low demand:

- Premiums are low, incentivizing agents to buy coverage
- LP yields are lower, which naturally limits excess capital

5.5 Three Operating Zones

Green Zone (0% to 60%): Abundant capital. Affordable premiums for agents. Moderate LP yield (Aave base + low premiums). Ideal conditions for buying insurance.

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Yellow Zone (60% to 80%): Growing demand. Premiums begin rising gradually. Balance between affordable coverage and growing LP returns. The market approaches the inflection point.

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Red Zone (80% to 95%): Scarce capital. Premiums surge exponentially. High LP yield attracts new depositors. Discourages new policies until liquidity returns. The protocol actively protects itself.

Blocked Zone (>95%): No new policies accepted. The vault protects itself from insolvency. Only existing policy execution and new deposits allowed.

5.6 Mathematical Solvency Guarantee

The model guarantees solvency through three mechanisms:

1. **1:1 Collateralization:** Each policy has dedicated USDC backing
2. **Exponential curve:** Premiums become prohibitive before reaching the limit, naturally discouraging over-issuance
3. **Absolute cap (95%):** Automatic rejection of new policies — it is impossible for the vault to accept more than it can cover

5.7 Impact Per Product

Product	P_base	Vault	Duration	Trigger
BTC Catastrophe Shield	1500 bps (15%)	Volatile Short/Long	7-30 days	>50% BTC drop
ETH Apocalypse Shield	2000 bps (20%)	Volatile Short/Long	7-30 days	>60% ETH drop
Depeg Shield	250 bps (2.5%)	Stable Short/Long	14-365 days	Stablecoin <\$0.95
IL Index Cover	850 bps (8.5%)	Volatile Short/Long	14-90 days	IL >2% at expiry
Exploit Shield	400 bps (4.0%)	Stable Long	90-365 days	Dual trigger (oracle+TEE)

All products share the same Kink Model (same M(U) curve) but differ in their P_base and risk factors.

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5.8 Protocol Fee Structure

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Event	Fee	Calculation	Recipient
Policy purchase	3%	3% of premium	Protocol (97% to vault)
Policy execution	3%	3% of payout	Protocol (97% to agent)
Vault withdrawal with profit	3%	3% of profit	Protocol

Complete example:

1. Agent buys BCS \$50,000, premium = \$280.82 - Protocol fee: $\$280.82 \times 3\% = \8.42
- Vault receives: \$272.40 2. Claim executes, payout = \$40,000 (80% coverage) -
Protocol fee: $\$40,000 \times 3\% = \$1,200$ - Agent receives: \$38,800 3. LP deposited
\$10,000, withdraws \$10,500 (profit \$500) - Performance fee: $\$500 \times 3\% = \15 - LP
receives: \$10,485 - If LP withdraws \$9,800 (loss): fee = \$0

Fee receiver: `0x2b4D825417f568231e809E31B9332ED146760337`

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6. Vaults and Yield

6.1 The Four Vaults

Lumina operates with four specialized vaults, each with a different cooldown period and assigned insurance products:

Vault	Cooldown	Assigned Products	Estimated APY	Address
VolatileShort	37 days	BCS + EAS + IL Index Cover	4 - 17%	<code>0xbd44...880d</code>
VolatileLong	97 days	Long IL + BCS/EAS overflow	4 - 21%	<code>0xFee5...f904</code>
StableShort	97 days	Short depeg	3 - 9%	<code>0x429b...54fC</code>
StableLong	372 days	Long depeg + Exploit Shield	3 - 10%	<code>0x1778...321c</code>

Vaults with longer cooldowns offer higher estimated APY because they assume greater risk (longer duration policies, less frequent but higher impact events).

6.2 Yield Composition

The yield that LPs receive comes from two sources:

- Aave V3 base yield (3-5% APY):** USDC deposited in the vaults is automatically deposited into Aave V3 Pool (`0xA238Dd80C259a72e81d7e4664a9801593F98d1c5`), generating passive yield in the form of aUSDC (`0x4e65fE4DbA92790696d040ac24Aa414708F5c0AB`).
- Insurance premiums:** When an agent purchases a policy, 97% of the premium (after the 3% protocol fee) is distributed to the corresponding vault, increasing the value of LP shares.

6.3 Soulbound Shares (Non-Transferable)

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Lumina's vault shares are **soulbound**: they cannot be transferred, sold, or used as collateral in other protocols. This design decision prevents:

- The creation of secondary markets for shares that could destabilize the vaults
- The use of shares as collateral in lending protocols, creating systemic risk
- The fragmentation of liquidity among multiple holders

The LP can only interact with their shares through `deposit()` and `withdraw()` (after the cooldown).

In future versions of the protocol, the implementation of an exclusive Lumina secondary market will be evaluated, where LPs can trade their positions. This market would operate under rules controlled by the protocol, maintaining vault stability while offering greater flexibility to participants.

6.4 Dedicated Collateral per Policy

Thanks to the 1:1 collateralization, each policy has its own dedicated backing in the vault. Even in the case of multiple simultaneous claims, each payment is guaranteed by already locked collateral. The vault priority system only applies to the allocation of new policies, not to the payment of existing claims.

7. Oracle and Verification

7.1 LuminaOracle

The LuminaOracle contract (`0x4d1140ac8f8cb9d4fb4f16cae9c9cba13c44bc87`) is the central data verification component of the protocol. It operates with an N-of-M multisig scheme that can be expanded according to decentralization needs.

It currently operates in 1-of-1 mode (expandable to N-of-M). The oracle verifies Chainlink feeds, implements a 1-hour sequencer check for Base L2, and validates data freshness before accepting it as input for triggers.

7.2 Chainlink Feeds Table

Feed	Address	Staleness	Products that use it
ETH/USD	<code>0x71041dddad3595F9CEd3DcCFBe3D1F4b0a16Bb70</code>	1,200s (20m)	BCS, EAS, IL Index Cover
BTC/USD	<code>0xCCADC697c55bbB68dc5bCdF8d3CBe83CdD4E071E</code>	1,200s (20m)	BCS, EAS
USDC/USD	<code>0x7e860098F58bBFC8648a4311b374B1D669a2bc6B</code>	86,400s (24h)	Internal reference
USDT/USD	<code>0xf19d560eB8d2ADf07BD6D13ed03e1D11215721F9</code>	86,400s (24h)	Depeg Shield
DAI/USD	<code>0x591e79239a7d679378eC8c847e5038150364C78F</code>	86,400s (24h)	Depeg Shield

Volatile asset feeds (ETH, BTC) have a staleness of 20 minutes, while stablecoin feeds have a staleness of 24 hours, reflecting the lower expected volatility.

7.3 Multisig Verification: verifyPackedMultisig

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The signature verification system uses a concatenated signature scheme ordered by address. Each authorized signer produces an ECDSA signature over the oracle data, and the signatures are concatenated in ascending order of signer address.

```
Signatures = signature_signer1 ++ signature_signer2 ++ ... ++ signature_signerN  
(where address(signer1) < address(signer2) < ... < address(signerN))
```

The `verifyPackedMultisig` function deserializes the signatures, verifies that each one comes from an authorized signer, that they are in the correct order (to prevent duplicates), and that the required N-of-M threshold is reached.

7.4 Phala TEE for Exploit Shield

The LuminaPhalaVerifier contract (`0x468b9D2E9043c80467B610bC290b698ae23adb9B`) uses Phala Network as a Trusted Execution Environment to verify the secondary trigger of Exploit Shield. The TEE executes verification logic in an isolated and tamper-proof environment, providing an independent security layer from the main oracle.

This is critical for Exploit Shield because exploits can involve oracle manipulation. By using an independent TEE, Lumina ensures that the trigger verification cannot be compromised by the same attack vector that caused the exploit.

8. Business Model

8.1 Monetization

Lumina Protocol generates revenue through a simple and transparent dual fee model:

Event	Fee	Description
Policy purchase (premium)	3%	3% of the premium goes to the protocol, 97% to the vault
Claim payout	3%	3% of the payout goes to the protocol, 97% to the agent
Vault withdrawal	3% performance	On the positive yield (profit over the original deposit)

8.2 Fee Receiver

All protocol fees are sent to the address:

Protocol Fee Receiver: 0x2b4D825417f568231e809E31B9332ED146760337

This address is controlled by the TimelockController with a 48-hour delay, ensuring transparency and auditability.

8.3 Performance Fee on Vault Withdrawals

A 3% performance fee is charged only on the positive yield (profit) when withdrawing funds from vaults. The fee is calculated on the difference between the withdrawn amount and the cost basis (original deposit). If there is no profit, no fee is charged.

Example: An LP deposits \$10,000 USDC. After accumulating yield, they withdraw \$10,500 USDC. The profit is \$500 (\$10,500 - \$10,000). The performance fee is $3\% \times \$500 = \15 . The LP receives a net of \$10,485.

This structure aligns the protocol's incentives with those of the LPs: Lumina only charges when the LP actually makes money.

8.4 Model Scalability

Lumina's business model is inherently scalable:

- **More LPs** deposit into vaults, increasing total coverage capacity
- **More capacity** allows issuing more policies and covering more agents
- **More policies** generate more premiums, which increases yield for LPs
- **Higher yield** attracts more LPs, closing the virtuous cycle

This flywheel is reinforced with each new participant, creating a positive network effect.

9. Security

9.1 Smart Contracts

The protocol's 13 contracts have been developed with Solidity security best practices:

- **Solidity 0.8.20:** Native overflow/underflow checking
- **CEI Pattern (Checks-Effects-Interactions):** All critical functions follow this pattern
- **SafeERC20:** All USDC transfers use OpenZeppelin's SafeERC20 library
- **ReentrancyGuard:** Protection against reentrancy attacks on all functions that handle funds
- **UUPS Proxies:** Upgradeable contracts with OpenZeppelin's UUPS pattern, controlled by TimelockController
- **119 tests:** Complete suite of unit and integration tests

9.2 Governance

The protocol's governance implements a layered security model:

- **TimelockController** (`0xd0De5D53dCA2D96cdE7FAf540BA3f3a44fdB747a`): 48-hour delay for all administrative operations. Any change to critical parameters requires a waiting period that allows LPs to react.
- **Gnosis Safe 1-of-1 (planned 2-of-3)** (`0xa17e8b7f985022BC3c607e9c4858A1C264b33cFD`): Currently 1-of-1 signer. Planned upgrade to require at least 2 of 3 authorized signers.

9.3 API and Backend

The API connecting AI agents to the contracts implements:

- **Rate limiting:** Request frequency control to prevent abuse
- **CORS:** Strict configuration of allowed origins
- **Helmet:** HTTP security headers
- **NonceManager:** Nonce management to prevent replay attacks and guarantee transaction sequentiality

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Oracle security is based on three layers:

- **Multisig:** N-of-M signature verification for oracle data
- **Chainlink TWAP:** Use of time-weighted average prices, not spot prices
- **Sequencer check:** Verification that the Base L2 sequencer has been active for at least 1 hour before accepting oracle data

9.5 Session Approval

For purchases made through relayers, the protocol requires **session approval**: the buyer must sign an explicit consent authorizing the relayer to execute the purchase on their behalf. This prevents a relayer from purchasing unauthorized policies with the user's funds.

9.6 OWS Integration

Lumina integrates the OWS (Open Wallet Standard) to facilitate secure interaction between AI agents and wallets. This integration allows agents to operate with granular and revocable permissions, without the need to expose private keys.

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10. Actuarial Data

10.1 BTC Catastrophe Shield / ETH Apocalypse Shield -- Expected Value for LPs

Metric	Value
Estimated annual premiums	~\$54,000
Estimated annual claims	~\$34,000
Annual net profit	+\$20,000
Margin	38%

The 38% margin reflects the fat-tail nature of BCS/EAS risk: catastrophic events are infrequent but severe. In years without events, the margin is significantly higher. In years with multiple events, the margin can be negative.

10.2 Depeg Shield -- Expected Value for LPs

Metric	Value
Estimated annual premiums	~\$85,000
Estimated annual claims	~\$32,000
Annual net profit	+\$52,000
Margin	62%

Depeg Shield has the highest premium volume and a healthy 62% margin. Depeg events are rare (1-2 times per decade for major stablecoins) but when they occur they are systemic.

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Metric	Value
Estimated annual premiums	Calculated by model
Annual net profit	+\$37,000
Margin	60%

IL Index Cover's proportional payout model and the 2% deductible create a favorable margin. Most policies expire with IL below 2%, generating premiums without claims.

10.4 Exploit Shield -- Expected Value for LPs

Metric	Value
Estimated annual premiums	Calculated by model
Annual net profit	+\$2,000
Margin	65%

Exploit Shield has the lowest absolute volume but the highest percentage margin. The dual trigger, the 14-day waiting period, and the \$50,000 per wallet cap significantly limit exposure.

10.5 Worst-Case Systemic Scenario

Metric	Value
Maximum estimated loss	-\$438,000
Percentage of total TVL (\$2M)	-21.9%
Estimated recovery time	10-12 months

A systemic scenario (market crash + depeg + exploit simultaneously) could generate a loss of 21.9% of TVL. Recovery is projected at 10-12 months through premium accumulation. This scenario is extremely unlikely given the independent correlation groups between products.

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11. Risks

Risk	Probability	Impact	Mitigation
Smart contract bug	Low	Critical	119 tests, CEI, ReentrancyGuard, Solidity 0.8.20, UUPS proxies for upgrades
Oracle manipulation	Low	High	TWAP (not spot price), multisig, sequencer check 1h, MAX_PROOF_AGE 30min
Flash loan attack	Low	High	TWAP multiple rounds, 1:1 collateral, no spot price dependency
Base L2 sequencer offline	Medium	Medium	1-hour sequencer check, policies do not expire during downtime
Stale Chainlink price	Medium	Medium	Staleness checks per feed (1200s volatile, 86400s stable), automatic rejection
Vault undercapitalization	Very Low	Critical	1:1 collateral, U_MAX 95%, automatic rejection of new policies upon reaching limit
Aave V3 exploit	Very Low	Critical	Accepted risk; Aave V3 is the most audited DeFi protocol; not covered (circular)
USDC depeg	Very Low	Critical	Accepted denomination risk; USDC not covered (circular)
Simultaneous multiple crash	Very Low	High	Vaults segregated by risk type, priority waterfall, correlation groups
Governance attack	Very Low	Critical	TimelockController 48h + Gnosis Safe 1-of-1 (planned 2-of-3), delay allows community reaction

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12. Roadmap

Phase 1 -- Launch (Current, Q1 2026)

- 5 operational insurance products: BCS, EAS, Depeg, IL Index, Exploit
- 4 liquidity vaults with Aave V3 yield
- Deployment on Base L2 (Chain 8453)
- 13 contracts in production with UUPS proxies
- Oracle with Chainlink feeds and Phala TEE support
- API for AI agents
- Governance via TimelockController + Gnosis Safe

Phase 2 -- Product Expansion (Q2-Q3 2026)

- **Gas Spike Shield:** Protection against extreme gas spikes
- **Slippage Shield:** Coverage against excessive slippage in trades
- **Bridge Shield:** Protection against cross-chain bridge failures
- New dedicated vaults for Phase 2 products
- Expansion of covered protocols in Exploit Shield

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Phase 3 -- Secondary Market and Token (Q4 2026 - Q1 2027)

- **Secondary market for policies and deposits (NFT marketplace):** Both active policies and vault deposit positions can be tokenized as NFTs and traded on a dedicated Lumina marketplace, allowing agents to transfer coverages and LPs to sell their positions before cooldown expiration.
- **Native token:** Launch of a utility and governance token for the protocol, with staking, voting, and premium discount features.

Phase 4 -- DAO and Multi-Chain (Q2-Q4 2027)

- **DAO governance:** Complete transition to decentralized governance by native token holders, replacing the Gnosis Safe model.
- **Cross-chain:** Expansion to Arbitrum and Optimism, enabling coverage of assets and protocols across multiple L2s.
- Integration with cross-chain communication protocols for unified settlement.

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13. Legal Framework

13.1 Protocol Structure

Lumina Protocol operates as a decentralized protocol deployed on Base L2 (Chain 8453). The smart contracts are immutable in their core logic, with upgrade capability through UUPS proxies controlled by a TimelockController with a 48-hour delay and a Gnosis Safe 1-of-1 (planned 2-of-3).

The protocol does not custody user funds. LP deposits are maintained in Aave V3, and claim payouts are executed directly from the vaults to the agents' wallets. The protocol only charges fees as an intermediary.

13.2 Product Nature

Lumina offers parametric protection products based on mathematically verifiable on-chain triggers. Lumina's products do not constitute insurance in the traditional regulatory sense: they do not require an insurer's license, do not involve subjective claim evaluation, and do not depend on judicial processes for dispute resolution.

Payments are automatic, deterministic, and verifiable by any third party that inspects the blockchain.

13.3 Regulatory Risks

The DeFi regulatory landscape continues to evolve. There are risks that future jurisdictions may classify on-chain parametric products as regulated insurance products, which could require licenses, additional regulatory compliance, or geographic restrictions.

The team actively monitors regulatory developments in major jurisdictions (US, EU, UK, Singapore) and is prepared to adapt the protocol's structure as necessary.

13.4 Disclaimer

THIS DOCUMENT IS FOR INFORMATIONAL PURPOSES ONLY AND DOES NOT CONSTITUTE FINANCIAL, LEGAL, OR INVESTMENT ADVICE. Participation in Lumina Protocol, whether as a coverage

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buyer or as a liquidity provider, involves significant risks including but not limited to: total loss of deposited capital, smart contract risks, oracle risks, regulatory risk, and market risk.

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Estimated yields (APY) are projections based on actuarial models and historical market conditions. They do not constitute a guarantee of future performance. Participants should conduct their own due diligence and consult with professional advisors before interacting with the protocol.

Lumina Protocol is provided "AS-IS" without warranties of any kind, express or implied.

14. Production Addresses

Network: Base L2 (Chain 8453) | **Deployment date:** March 29, 2026

Governance

Contract	Address
TimelockController	0xd0De5D53dCA2D96cdE7FAf540BA3f3a44fdB747a
Gnosis Safe 1-of-1 (planned 2-of-3)	0xa17e8b7f985022BC3c607e9c4858A1C264b33cFD

Core

Contract	Address
CoverRouter	0xd5f8678A0F2149B6342F9014CCe6d743234Ca025
PolicyManager	0xCCA07e0676222AA27DEd58482DeD3d9a7d0162a
LuminaOracle	0x4d1140ac8f8cb9d4fb4f16cae9c9cba13c44bc87
LuminaPhalaVerifier	0x468b9D2E9043c80467B610bC290b698ae23adb9B

Vaults

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VolatileShort	37 days	0xbd44547581b92805aAECc40EB2809352b9b2880d	
VolatileLong	97 days	0xFee5d6DAdA0A41407e9EA83d4F357DA6214Ff904	
StableShort	97 days	0x429b6d7d6a6d8A62F616598349Ef3C251e2d54fC	
StableLong	372 days	0x1778240E1d69BEBc8c0988BF1948336AA0Ea321c	

Shields (Products)

Shield	Address
BTC Catastrophe Shield (BCS)	0x36e37899D9D89bf367FA66da6e3CebC726Df4ce8
ETH Apocalypse Shield (EAS)	0xA755D134a0b2758E9b397E11E7132a243f672A3D
DepegShield	0x7578816a803d293bbb4dbea0efbed872842679d0
ILIndexCover	0x2ac0d2a9889a8a4143727a0240de3fed4650dd93
ExploitShield	0x9870830c615d1b9c53dfee4136c4792de395b7a1

External Contracts

Contract	Address
USDC (Circle)	0x833589fCD6eDb6E08f4c7C32D4f71b54bdA02913
Aave V3 Pool	0xA238Dd80C259a72e81d7e4664a9801593F98d1c5
aUSDC (Aave)	0x4e65fE4DbA92790696d040ac24Aa414708F5c0AB
Protocol Fee Receiver	0x2b4D825417f568231e809E31B9332ED146760337

Operational Keys

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Deployer / Owner	0xe585e76A0b8CbbC2d10b1110a9ac3F4c11dBfDa8	
Oracle Signer	0x933b15dd4F42bd2EE2794C1D188882aBCCDa977E	
Relayer	0xEdA7774A071a8DDa0c8c98037Cb542A1ee6aC7Eb	

Chainlink Feeds

Feed	Address	Staleness
ETH/USD	0x71041dddad3595F9CEd3DcCFBe3D1F4b0a16Bb70	1,200s (20m)
BTC/USD	0xCCADC697c55bbB68dc5bCdf8d3CBe83CdD4E071E	1,200s (20m)
USDC/USD	0x7e860098F58bBFC8648a4311b374B1D669a2bc6B	86,400s (24h)
USDT/USD	0xf19d560eB8d2ADf07BD6D13ed03e1D11215721F9	86,400s (24h)
DAI/USD	0x591e79239a7d679378eC8c847e5038150364C78F	86,400s (24h)

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15. Correlation Groups

Lumina's products are designed to cover risks with low correlation between them, limiting the possibility of simultaneous claims:

Correlation Group	Affected Products	Trigger	Joint Probability
ETH/BTC crash	BCS, EAS, IL Index Cover	Drop >30% in volatile assets	Medium
Stablecoin crisis	Depeg Shield	DAI or USDT depeg below \$0.95	Low
Protocol exploit	Exploit Shield	Hack or vulnerability in covered protocol	Low
Systemic event	BCS + EAS + Depeg + IL + Exploit	Complete DeFi ecosystem collapse	Very Low

Correlation Analysis

- **BCS, EAS and IL Index Cover** have high correlation: a market crash triggers BCS/EAS and simultaneously generates significant IL. For this reason, they share the VolatileShort vault.
- **Depeg Shield** has low correlation with volatility products. A stablecoin depeg can occur independently of market direction (e.g.: banking crisis, Tether regulatory risk).
- **Exploit Shield** has low correlation with all other products. Protocol hacks are idiosyncratic events that do not depend on market conditions.
- **Systemic event** (all triggers simultaneously) is the worst-case scenario. Vault segregation by risk type and the waterfall system partially mitigate this risk. The maximum estimated loss is -21.9% of TVL (\$438K out of \$2M).

16. API and Agents

16.1 General Description

The human, through their AI agent, interacts with Lumina Protocol through a REST API that exposes the protocol's core functionalities. The API allows quoting, purchasing policies, checking status, and verifying triggers without the need to interact directly with the contracts.

Main endpoints:

Endpoint	Method	Description
<code>/quote</code>	POST	Get a quote for a policy
<code>/purchase</code>	POST	Execute policy purchase via relay
<code>/policy/:id</code>	GET	Check policy status
<code>/vault/:name/stats</code>	GET	Get vault statistics
<code>/vault/:name/deposit</code>	POST	Deposit USDC into a vault
<code>/vault/:name/withdraw</code>	POST	Initiate withdrawal from a vault
<code>/oracle/price/:asset</code>	GET	Query current price of an asset

16.2 SKILL Reference

The complete API documentation and instructions for configuring agents can be found in the protocol's SKILL file: `LUMINA-SKILL.txt` (Version 3.0, March 2026).

The SKILL file contains:

- Detailed description of each product with parameters, examples, and pricing tables
- Step-by-step instructions for an agent to purchase coverage
- Instructions for depositing and withdrawing as an LP

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- Parameters for each vault and their estimated APYs
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- Fee information and how to plan coverages considering the 3% fee

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16.3 Agent Setup Flow

1. The human configures their agent with Lumina's API URL and their wallet credentials
2. The agent queries `/quote` with the desired parameters (product, asset, amount, duration)
3. The agent evaluates the quote and decides whether to proceed
4. The agent approves USDC to the CoverRouter via `approve()`
5. The agent executes `/purchase` or directly calls `CoverRouter.purchaseCover()`
6. The policy is registered on-chain in the PolicyManager
7. The agent can monitor their policy status via `/policy/:id`

17. Conclusion

Lumina Protocol represents the first parametric insurance infrastructure designed exclusively for the agentic economy. By combining verifiable mathematical triggers, decentralized oracles, strict 1:1 collateralization, and yield generated through Aave V3, the protocol offers a complete solution for the critical risks that AI agents face in DeFi.

Key Protocol Strengths

- **Pure parametric:** No subjective claims, no voting, no waiting. The trigger activates and the payout is immediate.
- **Designed for agents:** Programmatic API, session approval for relayers, OWS integration, SKILL documentation.
- **1:1 collateral:** Each policy is backed 100% by real USDC locked in the vault. No undercapitalization risk.
- **Real yield:** LPs earn compound yield from Aave V3 + insurance premiums, with estimated APYs of 3-21%.
- **Layered security:** Solidity 0.8.20, ReentrancyGuard, CEI, TimelockController 48h, Gnosis Safe 1-of-1 (planned 2-of-3), TWAP, sequencer check.
- **Transparent business model:** 3% premium + 3% payout. No hidden fees. 3% performance fee on positive yield from vault withdrawals.

Vision

Lumina aspires to become the standard insurance infrastructure for the agentic economy. As millions of AI agents manage trillions of dollars in DeFi assets, the need for programmatic, instant, and reliable coverage will be as fundamental as today's lending and trading infrastructure.

With the roadmap toward additional products (Gas Spike, Slippage, Bridge Shield), an NFT policy marketplace, native token, and DAO governance, Lumina positions itself as the reference protocol in decentralized insurance for the new generation of autonomous participants in the DeFi ecosystem.

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Lumina Protocol -- Parametric insurance for the agentic economy

Base L2 | USDC | Aave V3 | 2026